# Cybersecurity Implementation in OT System

***Network Security and Endpoint Protection***



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# Chapter 1: Introduction

## 1.1 Project Summary:

This project was developed as a practical demonstration to support my application for the role of **OT Cybersecurity Engineer**. It reflects my technical expertise and passion in applying cybersecurity principles to secure Operational Technology (OT) systems.

The Project focuses on building a comprehensive OT cybersecurity lab that simulates a real industrial environment. The lab integrates critical enterprise and control system components – including Active Directory Domain Service, Trellix ePolicy Orchestrator (ePO), Veritas Backup and Recovery, Window Server Update Services (WSUS), Foxboro DCS, and Modbus communication – to demonstrate secure architecture design, control implementations and risk mitigation strategies.

## 1.2 Scope:

The lab environment and documentation cover the following:

1. Execution of risk assessment, security control testing, and validation of system hardening.
2. Integration of enterprise and industrial components within a segmented network architecture (IT, DMZ, OT).
3. Implementation of Active Directory-based authentication and authorization.
4. Simulation of Foxboro DCS and Modbus communication to replicate OT data flow.
5. Deployment of firewalls and access control policies to enforce network segmentation and traffic filtering.
6. Deployment and configuration of Trellix Endpoint Security (ENS) and ePolicy Orchestrator (ePO) for endpoint protection and policy enforcement.
7. Implementation of Veritas Backup and Recovery for data protection.
8. Configuration of WSUS for centralized patch management.

## 1.3 Objectives:

The main objectives of the project are to:

1. Design and build a functional OT cybersecurity laboratory environment.
2. Conduct a risk assessment to identify potential threats to the lab environment.
3. Implement technical and administrative controls to mitigate identified risks.
4. Test and validate the effectiveness of implemented security measures.
5. Develop professional documentation to demonstrate cybersecurity design and engineering competency.

# Chapter 2: Risk Assessment

The objective of this phase is to identify and mitigate potential cybersecurity risks before implementation.

This Process is guided by the following standards and frameworks:

1. **NIST SP 800-30** – *Guide for Conducting Risk Assessments*
2. **ISA/IEC 62443-3-2** – *Risk Management Principle and Guidelines*

## 2.1 Methodology

The risk assessment follows a structured process consisting of the following steps:

1. System/Asset Characterization – Identify and document all critical assets.
2. Threat/Vulnerability Identification – Determine potential threat and weaknesses.
3. Risk Analysis – assess the likelihood and impact of identified risks.
4. Risk Evaluation – Prioritize risks based on severity.
5. Risk Treatment – Define and implement mitigation strategies.

## 2.2 Risk Rating Criteria

We will use a 5 x 5 Risk Matrix with two variables:

| Likelihood | Description |
|---|---|
| 1 | Rare – Unlike to occur |
| 2 | Unlikely – Could occur at some point |
| 3 | Possible – Might occur occasionally |
| 4 | Likely – Expect to occur in most condition |
| 5 | Almost Certain – Will occur frequently |

| Impact | Description |
|---|---|
| 1 | Insignificant– No disruption or data loss |
| 2 | Minor – Limited impact on non-critical system |
| 3 | Moderate – Temporary Service disruption |
| 4 | Major – Significant downtime or data compromise |
| 5 | Critical – Severe damage or control loss in OT System |

**Risk Rating = Likelihood x Impact**

| Risk Rating | Level | Action |
|---|---|---|
| 1-5 | Low | Acceptable, monitor only |
| 6-10 | Medium | Mitigate through standard control |
| 11-15 | High | Require immediate Mitigation |
| 16-25 | Critical | Unacceptable, redesigned required |

## 2.3 Asset Characterization

| Category | Asset | Description |
|---|---|---|
| IT System | Primary Domain Controller (PDC) Server | Central User Identification, group policy management and access control |
| | EPO Server | Centrally manage policies, monitor events and automate responses |
| | Window Server Update Services (WSUS) Server | Patch Management |
| | Backup Server (BAC1) | Provides data backup and recovery services |
| OT System | Engineering Workstation (10EWS1) Server | For Foxboro DCS Architecture, Includes Host for Foxboro Controller (FDC280), Galaxy Repository Server, CS (for Control Logic Editing, Visualization, etc...) |
| | Third Part Device (Such as PLC) – on Our Lab we will use Modbus Simulator | Connect Direct to the Field device and Send Data to the Foxboro DCS |
| Network Devices | Switch | For Device Connection and Network Segmentation. |
| | Firewall | Secure access between Third Part Devices and DCS. Also, between OT and IT System |

## 2.4 Threat/Vulnerability Identification, Risk Analysis and Risk Evaluation

| Risk ID | Threat/Vulnerability | Affected Asset | Likelihood | Impact | Risk Score | Risk Level |
|---|---|---|---|---|---|---|
| R1 | Unauthorized Access to Domain Controller and Other Server | All Server | 3 | 4 | 12 | High |
| R2 | Malware infection via USB or File Transfer | All Server | 3 | 5 | 15 | High |
| R3 | Network Misconfiguration | Switch, Firewall | 3 | 5 | 15 | High |
| R4 | Backup Corruption or Failed restore | Veritas Backup Server | 2 | 4 | 8 | Medium |
| R5 | Modbus Data Tempering | Modbus Devices, DCS | 2 | 5 | 10 | Medium |
| R6 | Misuse of Privileged account | PDC Server, Engineering Workstation (10EWS1) | 3 | 5 | 15 | High |
| R7 | Data Loss due to Hardware Fail | All Server | 3 | 5 | 15 | High |

## 2.5 Risk Treatment Plan

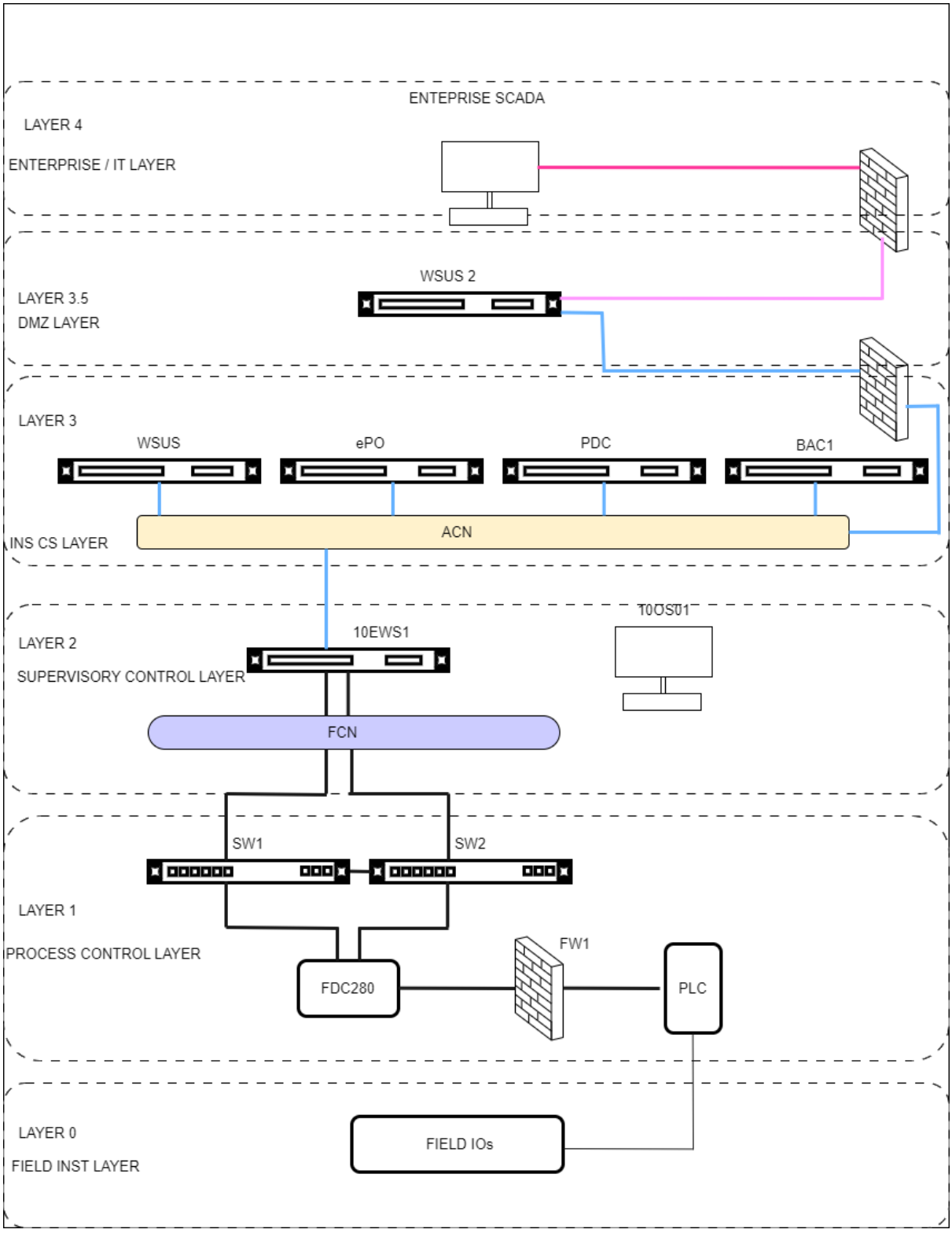
Each identified risks will be mitigated using appropriate technical, administrative, or procedure controls. The Implementation details are defined in **Chapter 4 – Control Implementation**.

| Risk ID | Treatment Strategy | Mitigation Measure | Responsible System/Tool |
|---|---|---|---|
| R1 | Prevent unauthorized Access to Domain and Other Server | Apply Strong Password Policy, restricted admin Group and enable auditing | Active Directory/GPO, EPO Server |
| R2 | Malware infection via USB or File Transfer | Enforce Endpoint Protection, Disable USB Ports (Unless for special permission) | PDC Server, EPO Server |
| R3 | Enforce Network Segmentation and Network Security | Implement Strict Firewall Policies, Use Segmentation (VLAN), and disable unused Switch/Firewall Ports | Switch and Firewall |
| R4 | Ensure Data Recovery | Schedule tests restore weekly, store backup on isolated storage | Backup (BAC1) Server |
| R5 | Secure Modbus Communication | Enable Firewall Filtering of Modbus Device | Firewall |
| R6 | Detect and monitor privilege abuse | Apply Least Privilege, log monitoring and periodic Access reviews | PDC Server, EPO Server |
| R7 | Maintain System Availability | Implement Veritas Backup, For AD DS – Also, Adding Secondary Domain Controller (SDC) | Backup (BAC1) Server |

# Chapter 3: System Design and Architecture

The architecture follows the ISA/IEC 62443 “Zone and Conduit” model, ensuring proper segregation between IT, DMZ, and OT layers while maintaining controlled and monitored communication between them.

## 3.1 Network Architecture



The Network Architecture is segmented intro three functional zones:

1. Enterprise (IT) Zone – Layer 4
2. DMZ Zone – Layer 3.5
3. OT Zone – Layer 0 to Layer 3

## 3.2 System Components, Functions and Data Flow

### *3.2.1 Primary Domain Controller (PDC) Server*

1. Utilizes Active Directory Domain Services (AD DS) for centralized user authentication, group policy management, and access control for all domain clients (EPO Server, 10EWS1 Server, WSUS Servers, and Backup Server).
2. Communicates with domain clients over the **Auxiliary Control Network (ACN).**

### *3.2.2 EPO Server*

1. Runs Trellix ePolicy Orchestrator (ePO) for centralized security policy management and monitoring of Trellix ENS clients across zones.
2. Communication with ENS clients occurs via the **ACN**.

### *3.2.3 10EWS1 Server*

This is the Engineering Workstation (Server) which will:

1. Hosts Foxboro DCS Controller (FDC280) and acts as the Galaxy Repository Server
2. Includes Foxboro Control Software for logic design and visualization using HMI.
3. Equipped with three NICs:
   a. Two NICs for **Foxboro Control Network (FCN)** communication with FDC280.
   b. One NIC for communication with PDC, EPO, and WSUS and BAC1 via **ACN**.

### *3.2.4 WSUS 2 Server*

1. Functions as the upstream Windows Server Update Services (WSUS) server, pulling security and system updates from Microsoft Update (Internet or corporate source).
2. Has two NICs:
   a. One for Internet access via **Enterprise Network (EN)**
   b. One for communication with WSUS Server via **ACN**.

### *3.2.5 WSUS Server*

1. Acts as the downstream WSUS server, receiving approved patches from WSUS 2 to prevent direct Internet connectivity between OT and IT layers.
2. Distributes patches to EPO, PDC, and Backup servers over **ACN**.

### *3.2.6 Backup Server*

1. Provides backup and restoration capabilities for critical servers and configurations.
2. Communicates with PDC, EPO, and 10EWS1 servers via **ACN**.

### *3.2.7 Field Device Controller (FDC280)*

1. Serves as the Foxboro DCS control processor, exchanging data with PLCs using Modbus TCP protocol.
2. Executes control logic to maintain process conditions.

3. Equipped with redundant NICs for **FCN** communication with the host server and additional two NICs for field device integration with third part devices using protocols such as Modbus TCP and OPC UA.

### *3.2.8 PLC*

1. Represents the field controller, communicating with FDC280 via Modbus TCP through the firewall.
2. Connects to physical field devices using I/O cables.

### *3.2.9 Switches*

1. Enable device connectivity and enforce network segmentation.

### *3.2.10 Firewall*

21. Segregates **IT**, **DMZ**, and **OT zones**, enforcing traffic filtering and logging for secure communication.

| Server name | Description | IP address | Subnet | Default Gateway | DNS |
|---|---|---|---|---|---|
| PDC | PDC – ACN Network | 192.168.40.10 | 255.255.255.0 | 192.168.40.100 | 127.0.0.1 |
| SDC | SDC – ACN Network | 192.168.40.20 | 255.255.255.0 | 192.168.40.100 | 192.168.40.10 |
| 10EWS1 | MESH A – FCN Network | | | | |
| | MESH B – FCN Network | | | | |
| | 10EWS1 – ACN | 192.168.40.30 | 255.255.255.0 | 192.168.40.100 | 192.168.40.10 |
| | ENET PORT | 192.168.120.20 | 255.255.255.0 | 192.168.120.100 | |
| EPO | EPO – ACN Network | 192.168.40.40 | 255.255.255.0 | 192.168.40.100 | 192.168.40.10 |
| WSUS | WSUS – ACN Network | 192.168.40.50 | 255.255.255.0 | 192.168.40.100 | 192.168.40.10 |
| WSUS 2 | ACN Network | 192.168.40.60 | 255.255.255.0 | 192.168.40.100 | 192.168.40.10 |
| | Enterprise Network (EN) | | | | |
| BAC1 | ACN Network | 192.168.40.70 | 255.255.255.0 | 192.168.40.100 | 192.168.40.10 |
| PLC | MODSIM PORT | 192.168.150.1 | 255.255.255.0 | 192.168.150.100 | |

# Chapter 4: Control Implementation

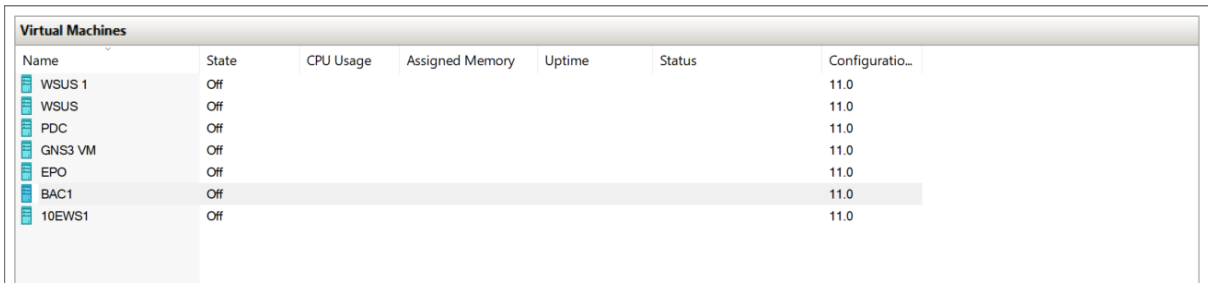
In this Chapter we will apply defence-in-depth principles for practical cybersecurity Control that protect OT System from cyber threat. The lab environment is based on a Foxboro DCS System Architecture.

## 4.1 Scope of the implementation:

The following activities are performed in this section:

1. Create Virtual Machines: PDC Server, EPO Server, 10EWS1 Server, BAC1 Server, WSUS Server and WSUS1 Server
2. Creating Virtual Switches: MESH A, MESH B, ACN, EN
3. Configuring PDC Server
4. Configuring 10EWS1 Server
5. Configuring EPO Server
6. Configuring BAC1 Server

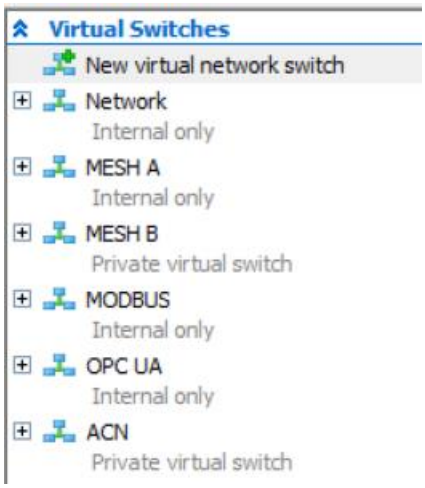
### *4.1.1 Virtual Machine Creation*



Virtual Machines

| Name | State | CPU Usage | Assigned Memory | Uptime | Status | Configuratio... |
|---|---|---|---|---|---|---|
| WSUS 1 | Off | | | | | 11.0 |
| WSUS | Off | | | | | 11.0 |
| PDC | Off | | | | | 11.0 |
| GNS3 VM | Off | | | | | 11.0 |
| EPO | Off | | | | | 11.0 |
| BAC1 | Off | | | | | 11.0 |
| 10EWS1 | Off | | | | | 11.0 |

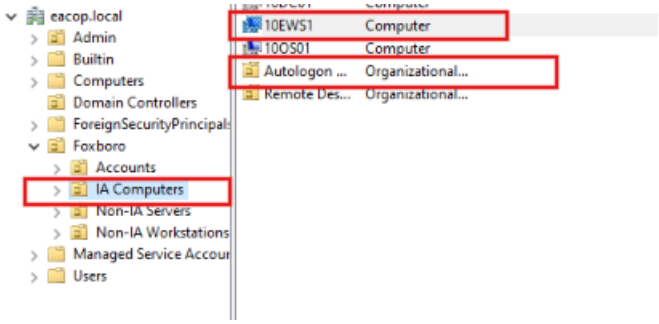
### *4.1.2 Creating Virtual Switches*



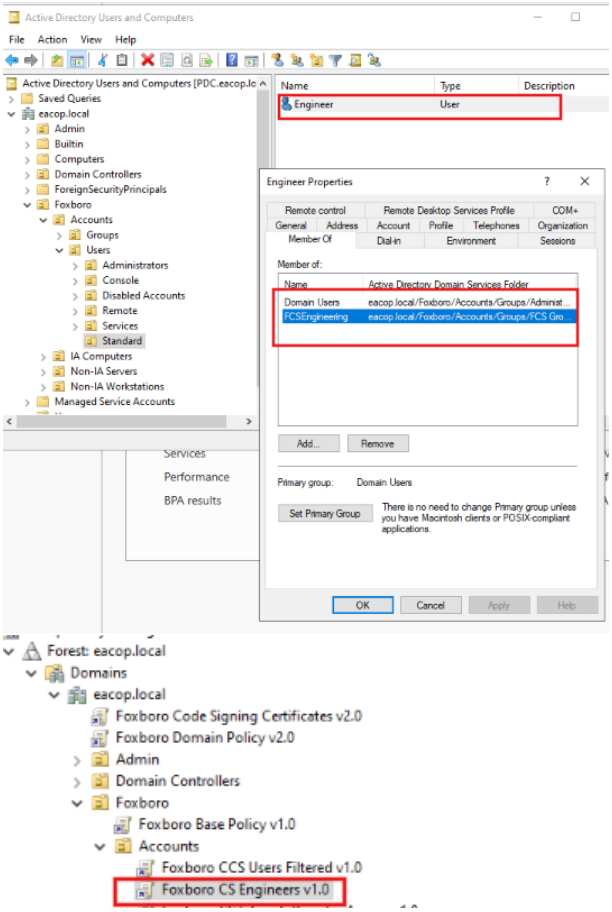

### *4.1.3 Configuring PDC Server*

1. Assign ACN Virtual Switch to PDC VM.
2. Installing Window server 2022 to the PDC VM.
3. Rename the server's name to PDC.
4. Assign the IP address, subnet, default gateway of the ACN virtual adapter as described to network description.
5. Install LGPO.
6. Install Trellix ENS.

7. Run full scan of the system to confirm no virus are present.
8. Disable Trellix Endpoint Security's "Scan trusted installers" feature.
9. Install Enterprise Edition CCS v9.8 for New Off-Control Domain Controller.
10. Enable Trellix Endpoint Security's "Scan trusted installers" feature.
11. On a Domain Controller's **IA Computers** OU, select 10EWS1 and drag it to the **Autologon Consoles** folder.



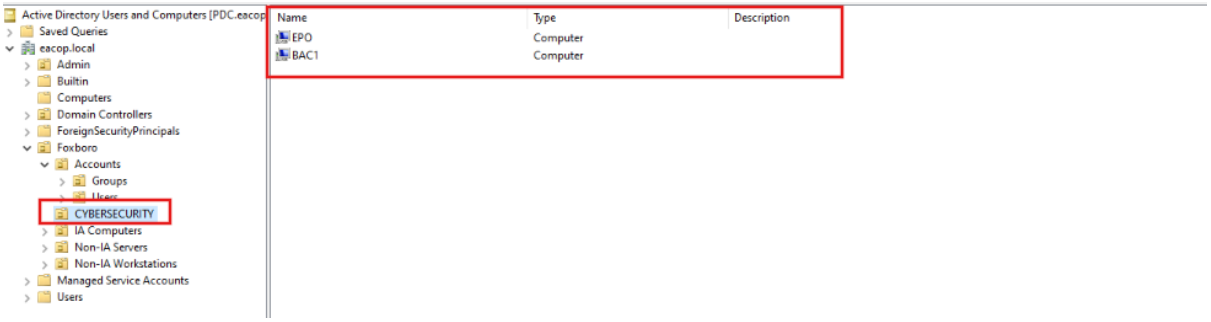

12. Adding 10EWS1 to DNS forward lookup table.
13. Create user Engineer and add to FCSEngineering Group.







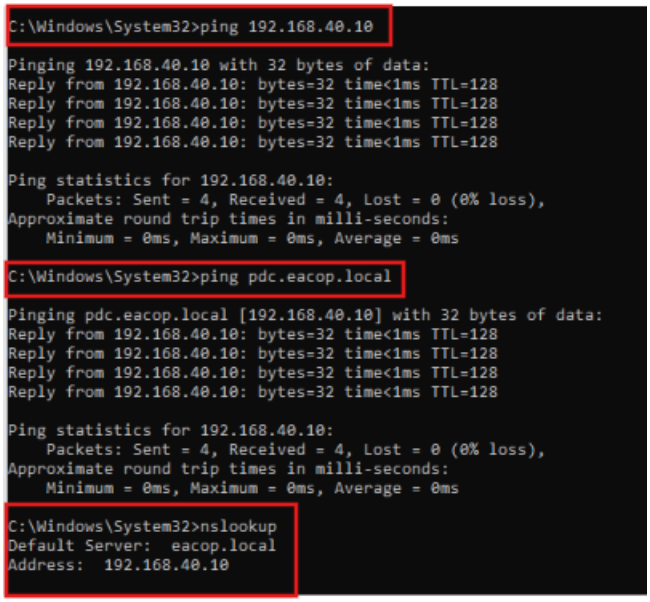
14. Create an Organization Unit under **Foxboro OU** and name as **CYBERSECURITY** and add EPO Server, and BAC1 Server.




15. Install CS on a PDC and create control software network account.

### *4.1.4 Configuring 10EWS1 Server*

1. Assigning ACN, MESH A, MESH B Virtual switches to 10EWS1 VM
2. Installing Window server 2022 to the 10EWS1 VM
3. Rename the server's name to be the same as defined on system definition (10EWS1)
4. Assign the IP address, subnet, default gateway of the ACN, MESH A, MESH B virtual adapter as described to network description.
5. Adding 10EWS1 IP address to the forward lookup zone on a PDC
6. Verify you can ping PDC using 192.168.40.10.
7. Verify you can ping PDC using FQDN: pdc.eacop.local
8. Verify the nslookup command shows the PDC's FQDN or it's IP address.



```
C:\Windows\System32>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Windows\System32>ping pdc.eacop.local

Pinging pdc.eacop.local [192.168.40.10] with 32 bytes of data:
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Windows\System32>nslookup
Default Server:  eacop.local
Address:  192.168.40.10
```

9. Set the ACN IP to Automatic (use DHCP) and obtain DNS automatic.
10. Install LGPO.
11. Install Trellix ENS.
12. Run full scan of the system to confirm no virus are present.
13. Disable Trellix Endpoint Security's "Scan trusted installers" feature.
14. Install Enterprise Edition CCS v9.8 for New Off-Control Domain Client.
15. Installing System Manager.
16. Installing Control Software V8.0.

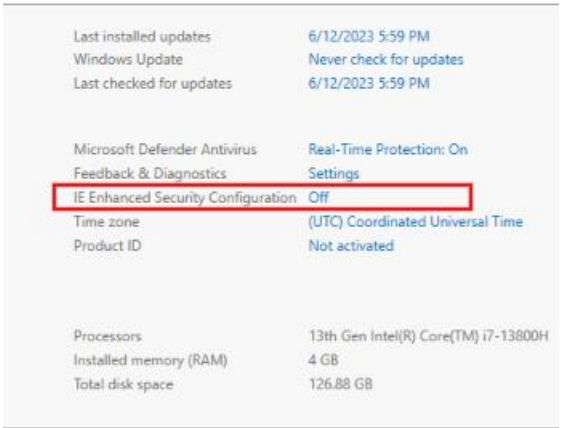
### *4.1.5 Configuring BAC1 Server*

6. Assign ACN Virtual Switch to BAC1 VM

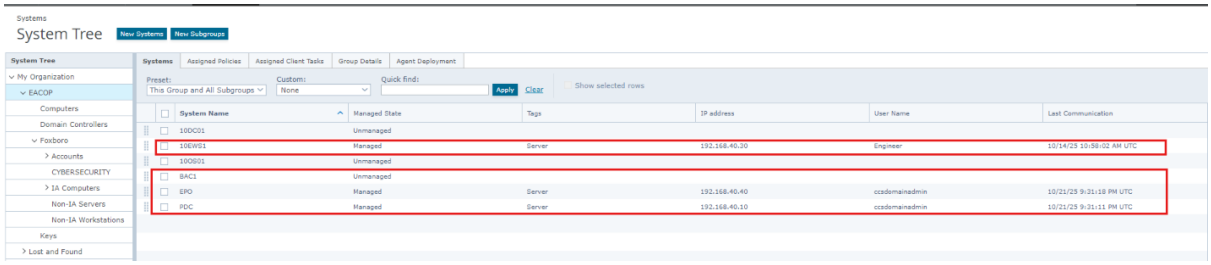
7. Installing Window server 2022 to the BAC1 VM
8. Rename the server's name to be BAC1
9. Assign the IP address, subnet, default gateway of the ACN virtual adapter as described to network description.
10. Add BAC1 server to the Domain Controller and login as CCS Domain Admin Account.
11. Install Trellix ENS.

### *4.1.6 Configuring EPO Server*

11. Assign ACN Virtual Switch to EPO VM
12. Installing Window server 2022 to the EPO VM
13. Rename the server's name to be EPO
14. Assign the IP address, subnet, default gateway of the ACN virtual adapter as described to network description.
15. Add EPO server to the Domain Controller and login as CCS Domain Admin Account.
16. Install Trellix ENS
17. Ensure IE ESC is off for both administrator and users




18. Install SQL server express 2022
19. Install ePolicy Orchestrator
20. Set **IE ESC** is on for both administrator and users
21. Deploy Trellix agent to PDC Server, EPO Server, 10EWS1 Server. We will not Deploy Trellix Agent to BAC1 Server as it will act as a Rogue System for Rogues System System.



# Chapter 5: Risk Treatment and Validation

This chapter outlines the treatment strategies applied to mitigate the risks identified during the risk assessment phase (Chapter 2).

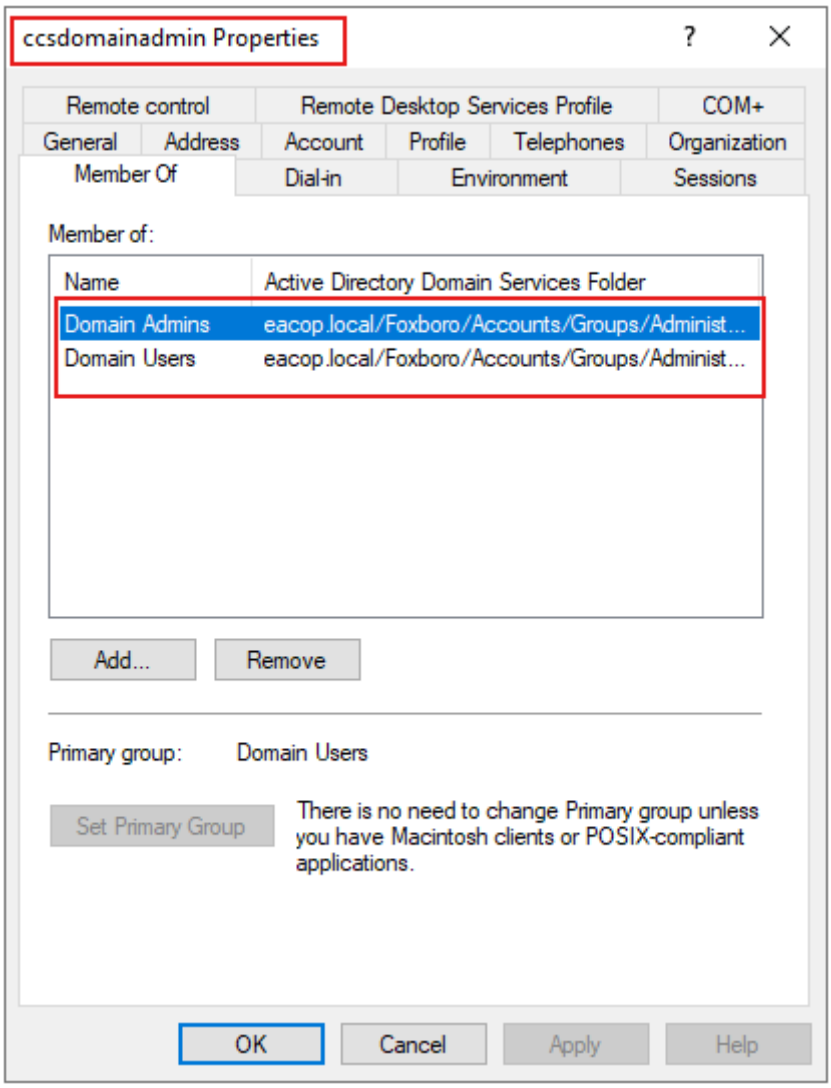
## 5.1 R1 & R6 – Prevent Unauthorized Access to PDC & Other Servers

The PDC stores all credentials, security Policies, and access permission for all connected Servers. If compromised, an attacker can control the entire OT system.

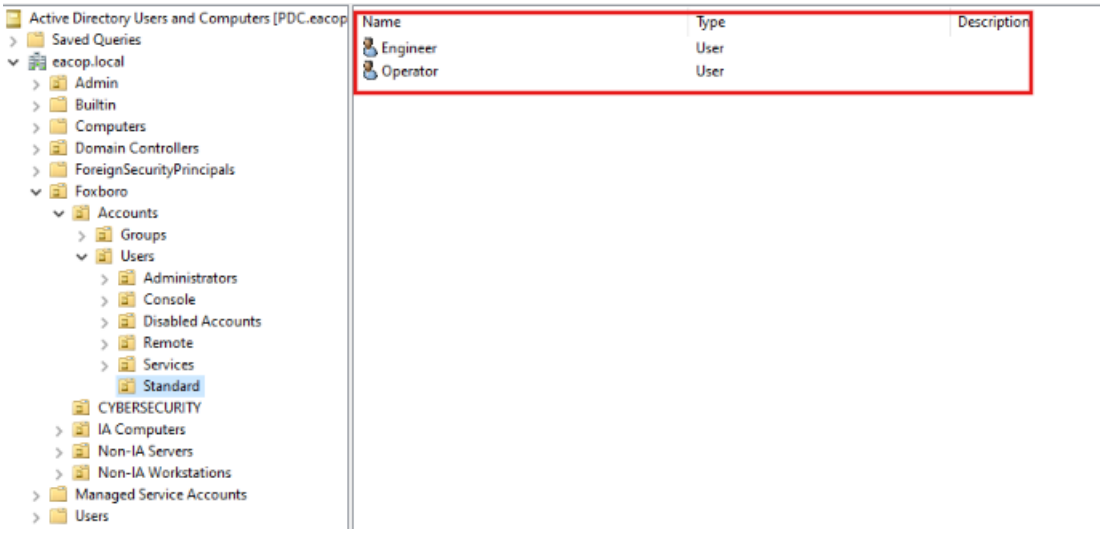
Therefore, PDC protection must combine technical, administrative, and physical layer of security.

### *5.1.1 Privileged Account Separation and Role Based Access Control*

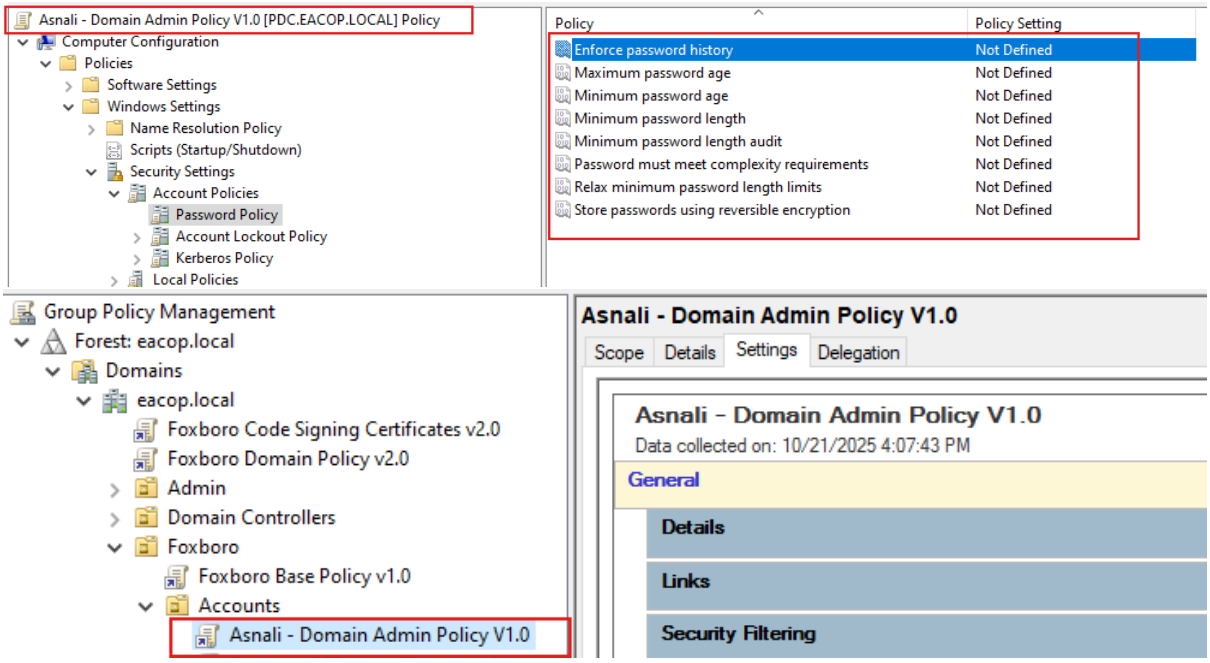
1. Create a **Domain Admin** group and a **Domain Admin** user, then assign the user as a member of the Domain Admin group.



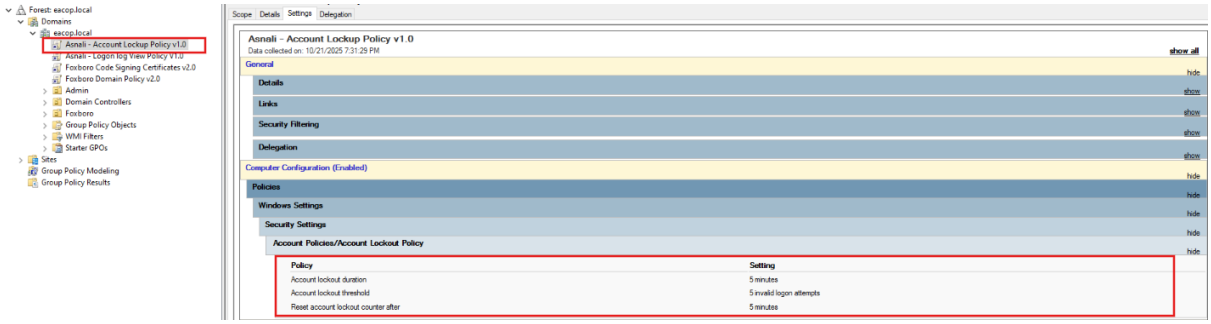
2. Create two user accounts: **Engineer** and **Operator**. Assign the Engineer account to the **IA Plant Engineer** group and the Operator account to the **IA Plant Operator** group. The Operator will have **view-only access to the HMI** and will not be permitted to edit control logic on the 10EWS1 Engineering Workstation



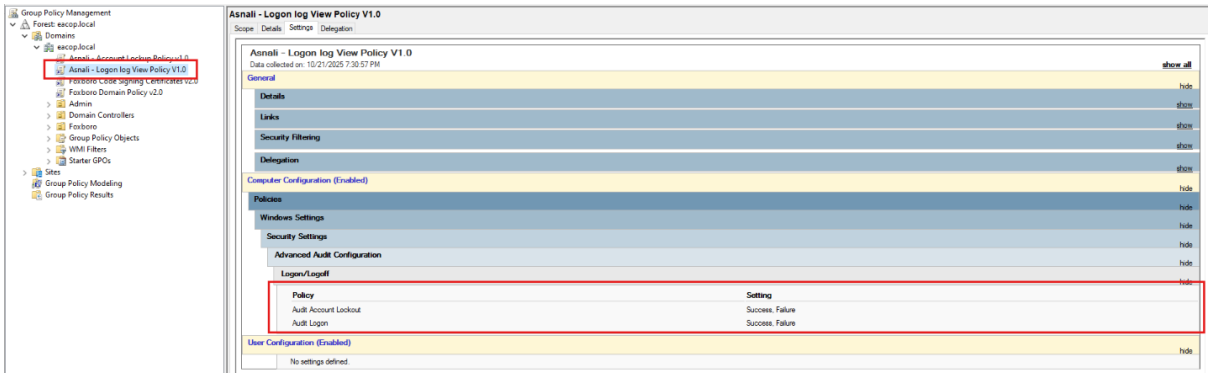

### *5.1.2 Apply Strong Password Policy: On this Lab we will not define Password Policy*



### *5.1.3 Account Lockup Policy*

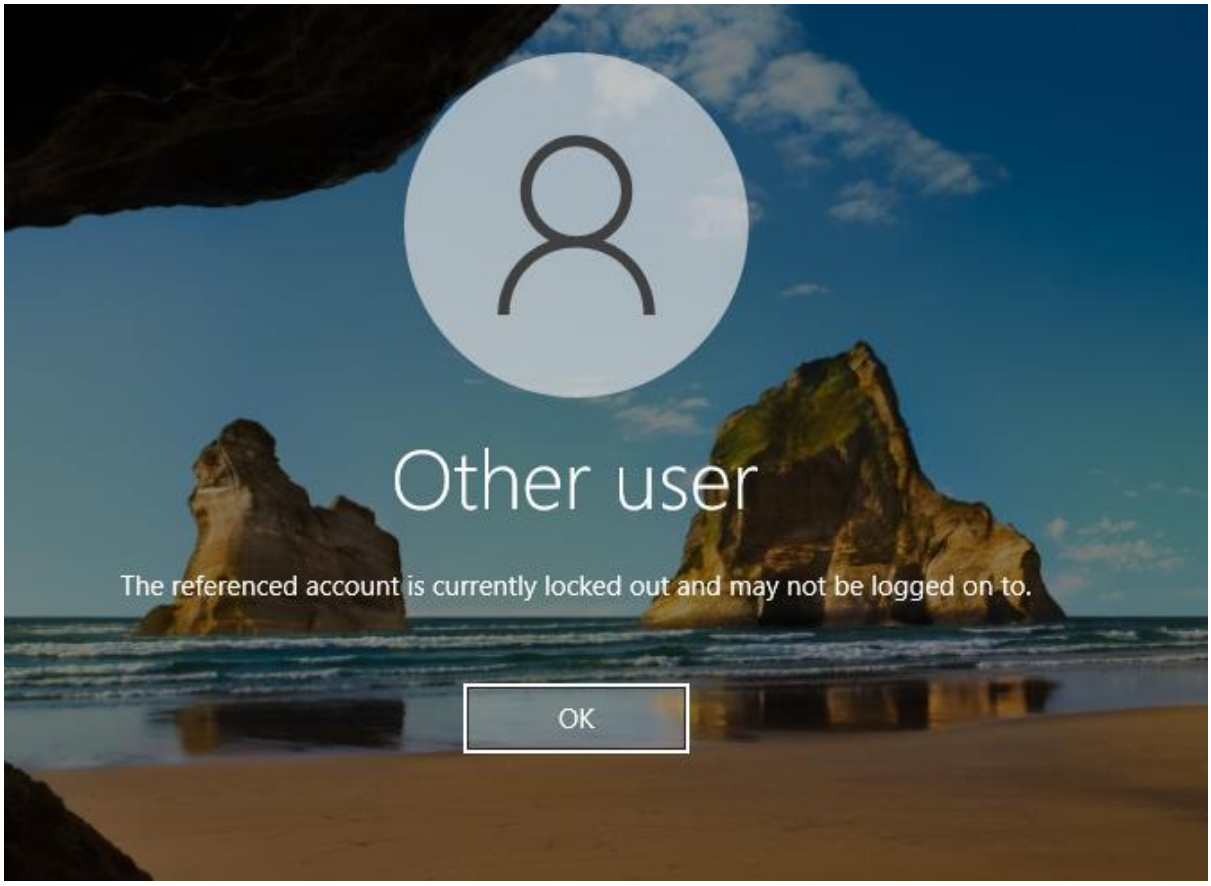



### *5.1.4 Enable detailed Login attempt on PDC Server and EPO Server*

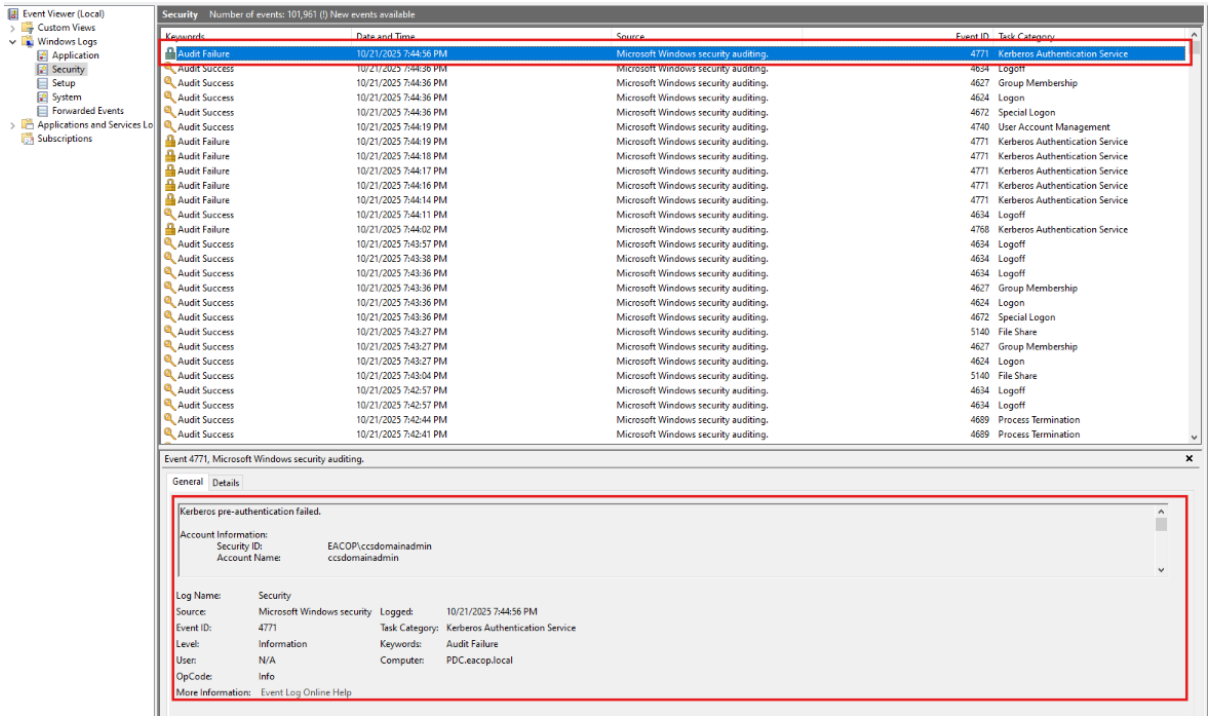



### *5.1.5* **Validation***:*

1. Perform a **failed login attempt** on the EPO Server to validate authentication and logging controls.



2. After attempt of false login: We can the Event ID on a PDC Event Viewer *4771 – Kerbos Authentication Service*

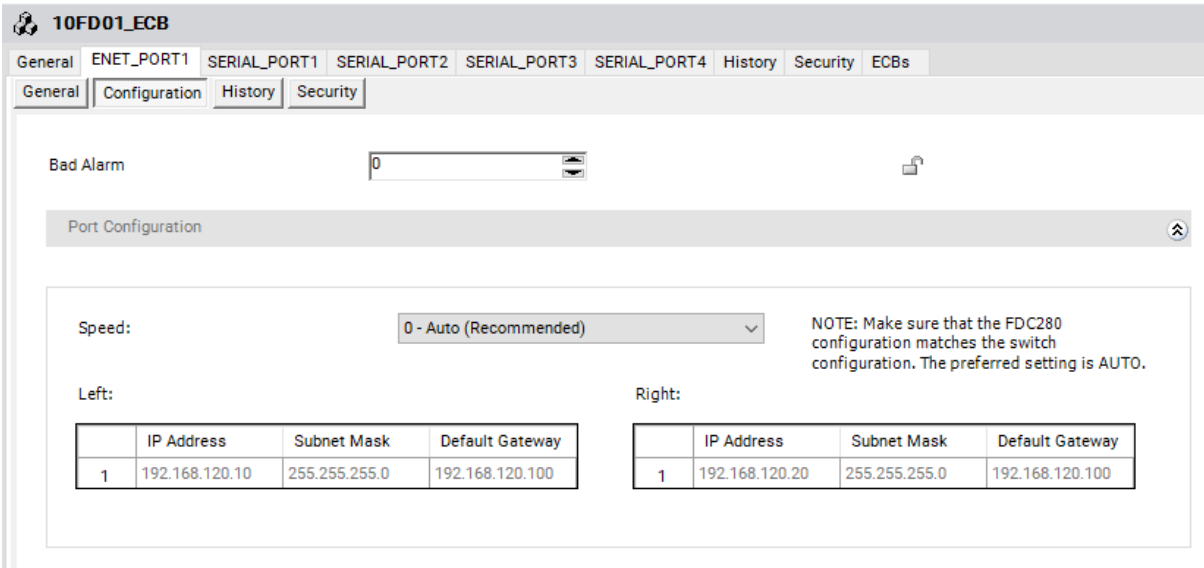



## 5.2 R5 – Network Segmentation and Security

In this lab, any third-party device connected to the Foxboro DCS system must first pass through a firewall for security filtering. This ensures controlled communication between the DCS and external devices.

### *5.2.1 Foxboro DCS (FDC280) Network Configuration*

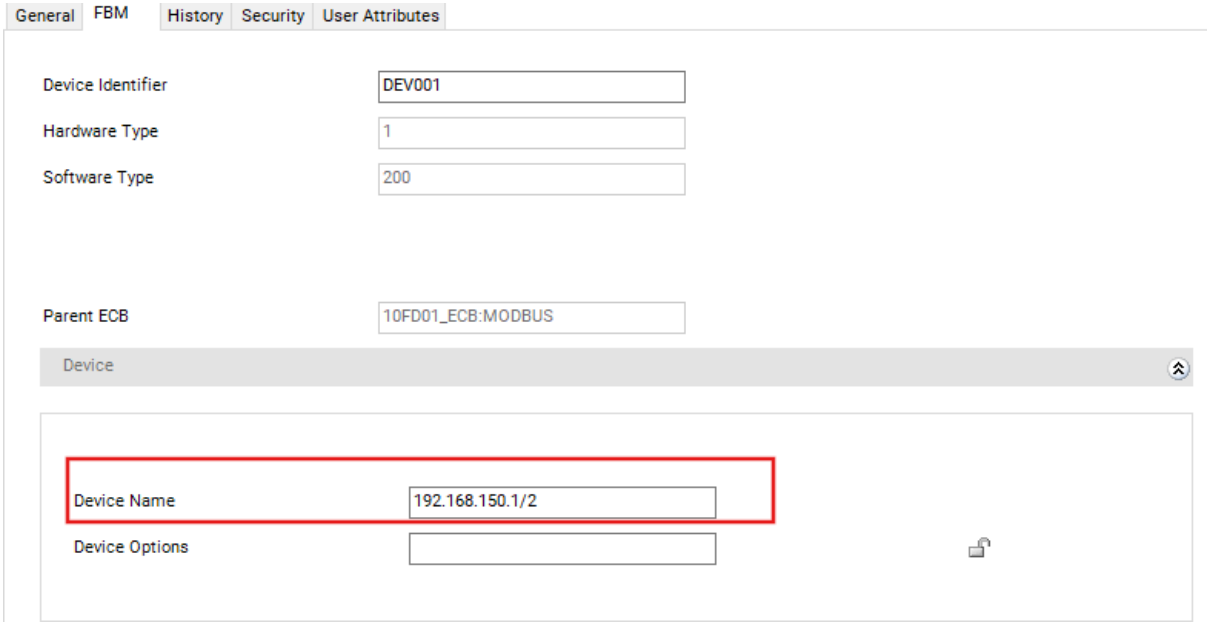
3. Configure DCS Ethernet ports




| | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|
| 1 | 192.168.120.10 | 255.255.255.0 | 192.168.120.100 |

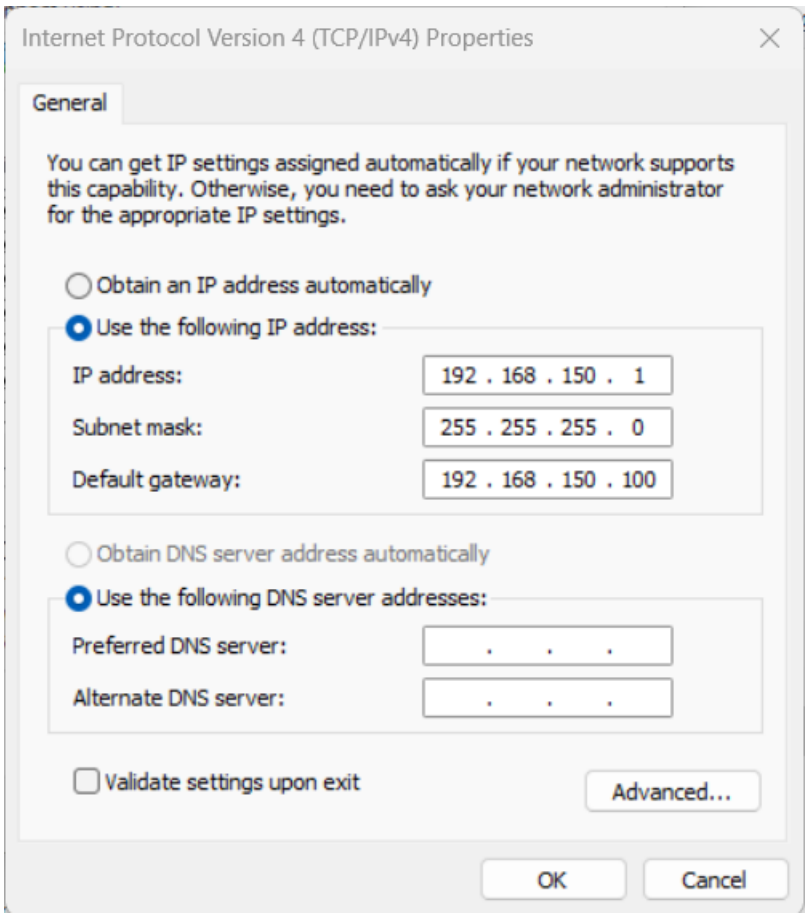
| | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|
| 1 | 192.168.120.20 | 255.255.255.0 | 192.168.120.100 |

4. Configure Device port for data exchange with PLC using Modbus TCP Protocol




### *5.2.2 Modbus Simulator Network Configuration*

Assign IP addresses for the Modbus Simulator to enable communication with the DCS through the firewall

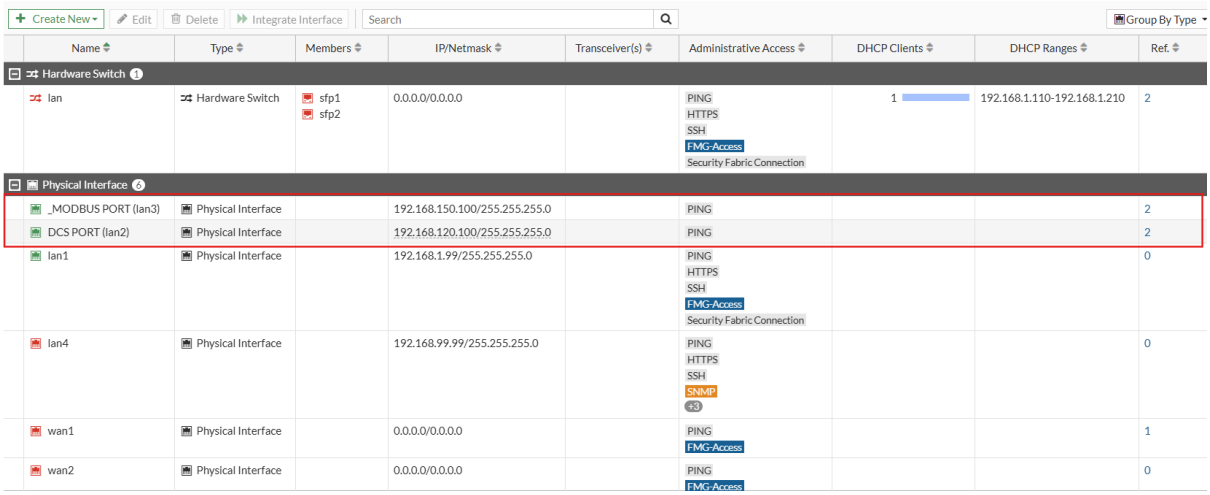


### *5.2.3 Firewall Policy Configuration*

The Modbus network and DCS network operate on **separate network segments**. Therefore, the firewall will use two distinct physical interfaces to manage traffic between them.

**Lab Steps:**

1. Configure physical interfaces for both networks on the firewall.



| Name | Type | Members | IP/Netmask | Transceiver(s) | Administrative Access | DHCP Clients | DHCP Ranges | Ref. |
|---|---|---|---|---|---|---|---|---|
| **Hardware Switch** | | | | | | | | |
| lan | Hardware Switch | sfp1<br>sfp2 | 0.0.0.0/0.0.0.0 | | PING<br>HTTPS<br>SSH<br>FMG-Access<br>Security Fabric Connection | 1 | 192.168.1.110-192.168.1.210 | 2 |
| **Physical Interface** | | | | | | | | |
| _MODBUS PORT (lan3) | Physical Interface | | 192.168.150.100/255.255.255.0 | | PING | | | 2 |
| DCS PORT (lan2) | Physical Interface | | 192.168.120.100/255.255.255.0 | | PING | | | 2 |
| lan1 | Physical Interface | | 192.168.1.99/255.255.255.0 | | PING<br>HTTPS<br>SSH<br>FMG-Access<br>Security Fabric Connection | | | 0 |
| lan4 | Physical Interface | | 192.168.99.99/255.255.255.0 | | PING<br>HTTPS<br>SSH<br>SNMP<br>+3 | | | 0 |
| wan1 | Physical Interface | | 0.0.0.0/0.0.0.0 | | PING<br>FMG-Access | | | 1 |
| wan2 | Physical Interface | | 0.0.0.0/0.0.0.0 | | PING<br>FMG-Access | | | 0 |

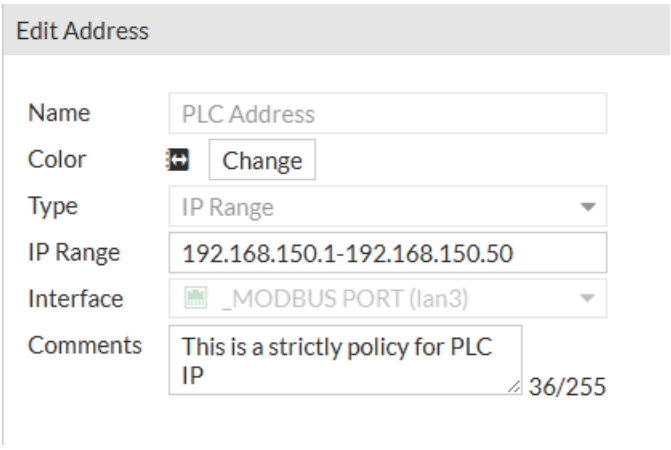
5. Create **two security policies** to allow controlled communication between the DCS and Modbus device (PLC)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| _MODBUS PORT (lan3) → DCS PORT (lan2) | | | | | | | | |
| MODBUS TO DCS | PLC Address | DCS IP | always | Modbus Port | ACCEPT | Enabled | no-inspection | UTM | 480 B |
| DCS PORT (lan2) → _MODBUS PORT (lan3) | | | | | | | | |
| DCS TO MODBUS | DCS IP | PLC Address | always | Modbus Port | ACCEPT | Enabled | no-inspection | UTM | 196.49 kB |

a. **Policy 1: Modbus –> DCS**
Allows communication from the PLC (source IP) to the DCS (destination IP) using **service port 502** (Modbus TCP).

f. **Policy 2: DCS –> Modbus**
Allows communication from the DCS (source IP) to the PLC (destination IP) using the same service port.

**PLC Address**



**DCS Address**

| | |
|---|---|
| Name | DCS IP |
| Color | Change |
| Type | IP Range |
| IP Range | 192.168.120.1-192.168.120.10 |
| Interface | DCS PORT (lan2) |
| Comments | Write a comment... 0/255 |

**Service Port**

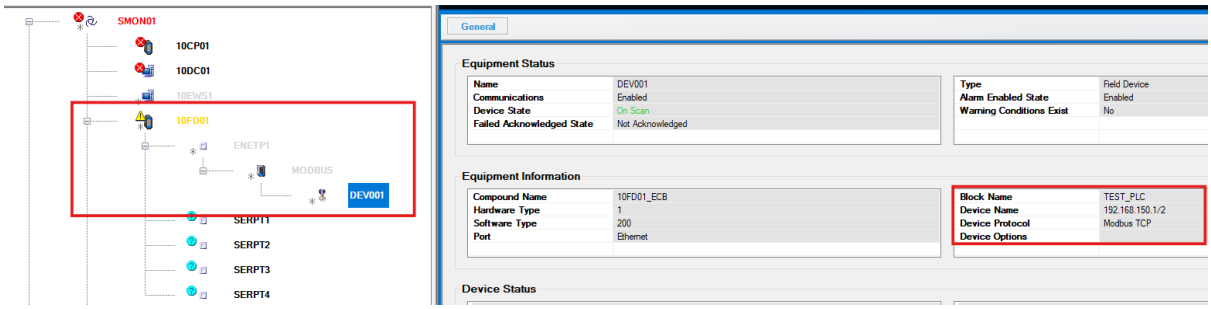
| | |
|---|---|
| Name | Modbus Port |
| Comments | Write a comment... 0/255 |
| Color | Change |
| Category | Uncategorized |

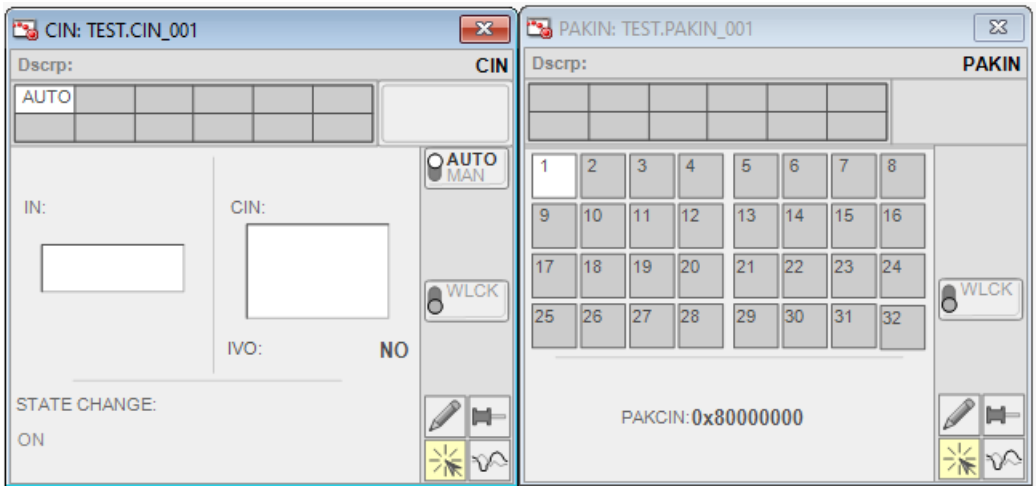
Protocol Options

| | |
|---|---|
| Protocol Type | TCP/UDP/SCTP |
| Address | IP Range FQDN 0.0.0.0 |
| Destination Port | TCP 502 - High |
| Specify Source Ports | |

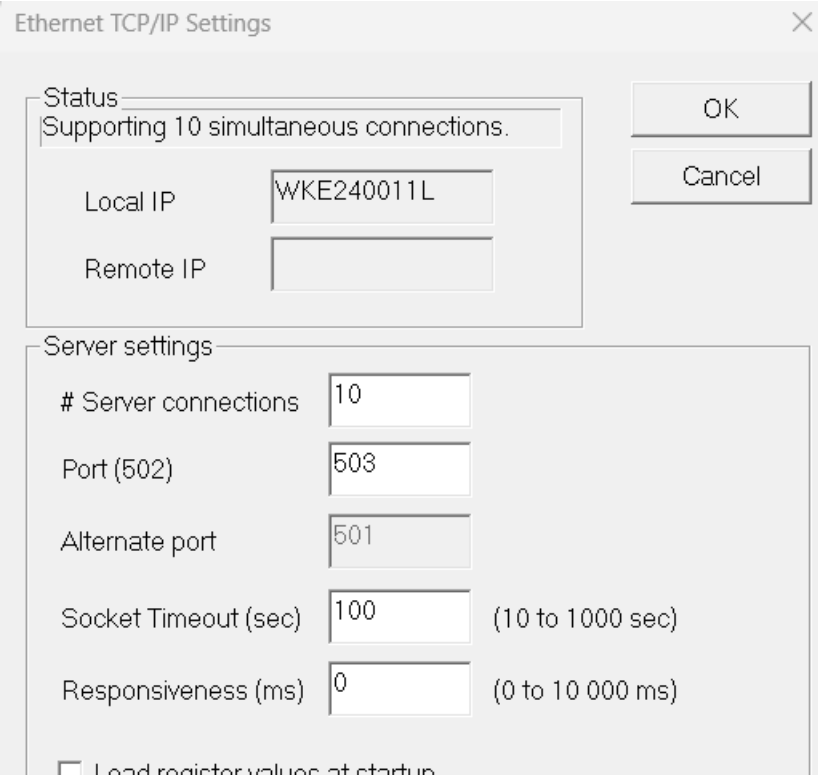
### *5.2.4 Validation:*

1. Confirm active communication by monitoring data exchange (e.g., receiving a value of **1** from the PLC on the first bit)

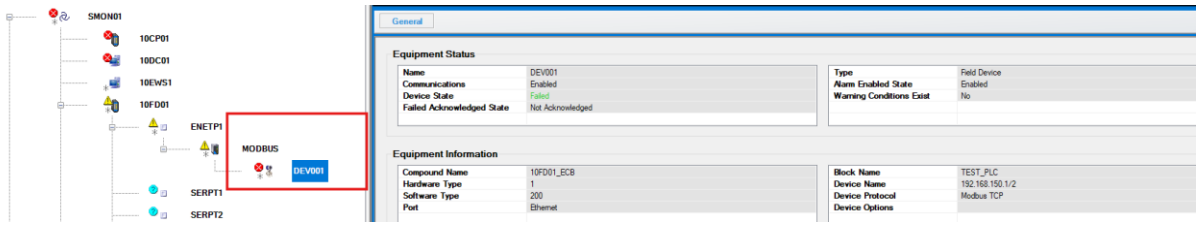


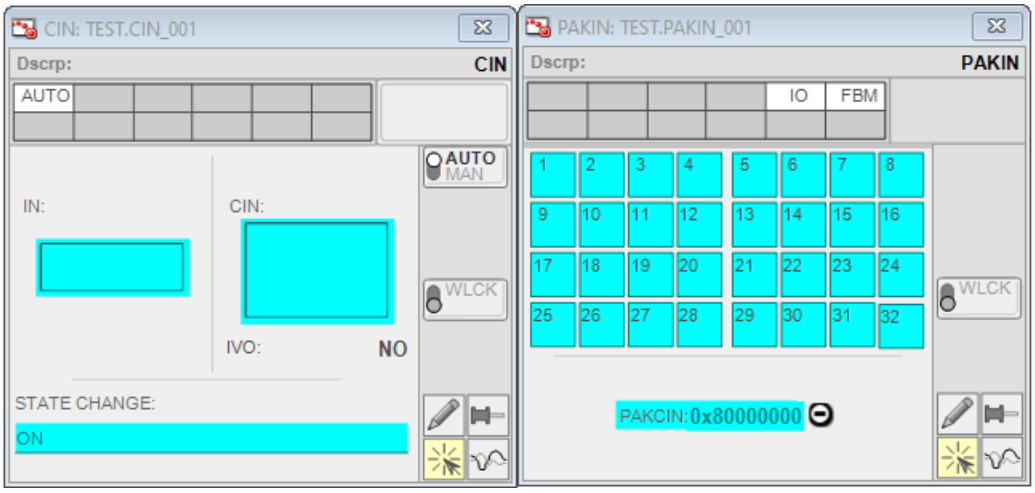



2. Change the Modbus service port to **503** and verify that communication is blocked, confirming the firewall policy enforcement.




Confirm the communication is disabled.






## 5.3 R2 – Endpoint Protection

Trellix Endpoint Security (ENS) provides real-time protection, behavioural analysis and exploit prevention on every endpoint (Servers).

All ENS Clients are managed by Trellix ePolicy Orchestrator (ePO), which provides centralized policy management, deployment and monitoring.

### *5.3.1 Rogue System Detection*

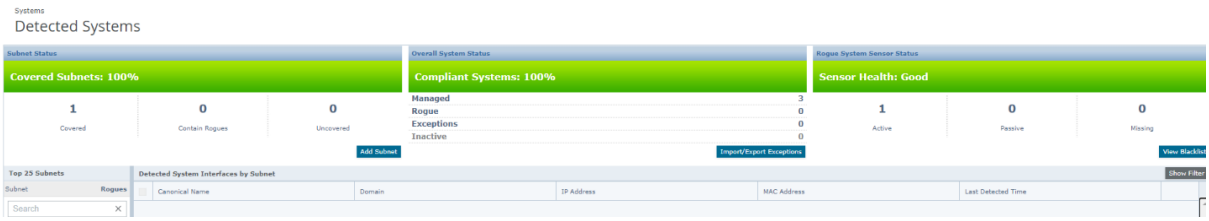
This is a tool that scans network to find the devices that are: -

1. Do not have the Trellix Agent installed
2. Are not managed by ePO
3. Might pose a security risk

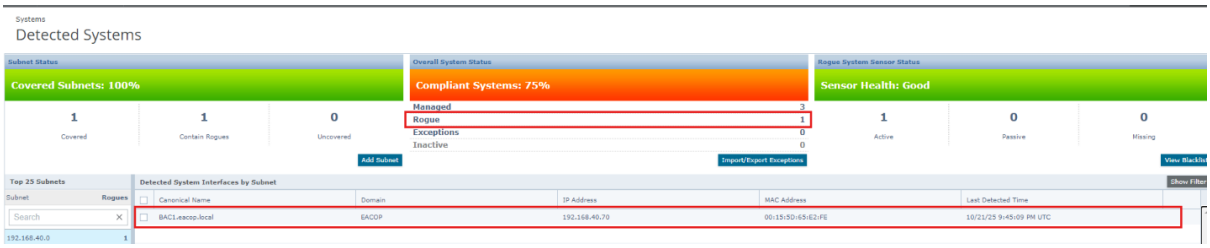
**Lab Steps**

1. Deploy Trellix Agent on PDC Server, ePO Server, and 10EWS1 Server.
2. We will not deploy Trellix Agent on BAC1 Server (to simulate a rogue system)
3. Install Rogue System Detection Extension and Check in RSD Package.
4. Install Rogue Sensor on ePO Server
5. We will connect PDC Server to a WSUS Server to Simulate a rogue system.

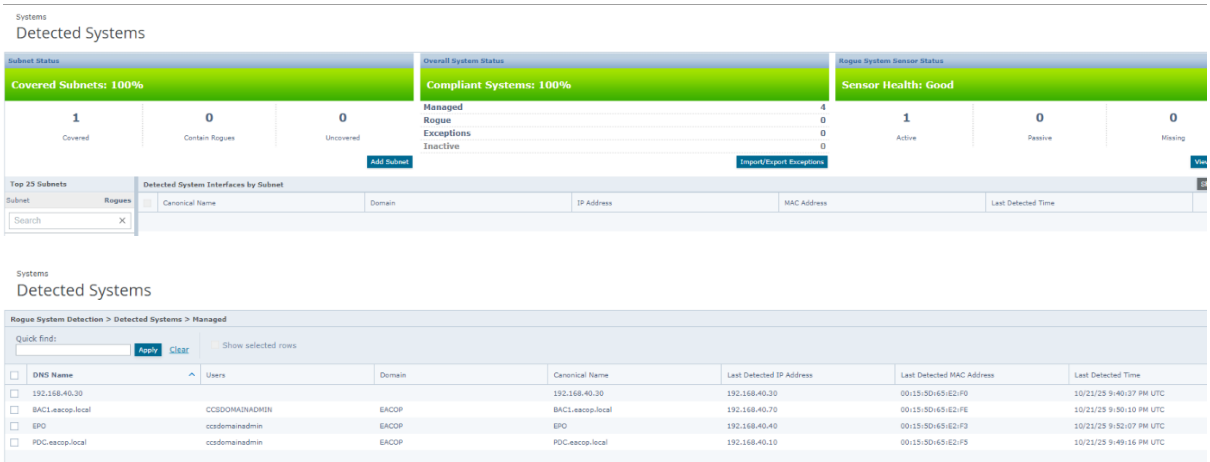
On below image the BAC1 is not added to the network – you can see system compliant is 100%



Add BAC1 Server in the ACN Network and verify that it is detected as a **rogue system**




Deploy the Trellix Agent to the BAC1 server from EPO Server to bring it under centralized management and enforce security poilicies.




### *5.3.2 Application, Change, and Integrity Control using Solidcore*

This ensures system integrity by preventing unauthorized modifications to endpoints and configurations.

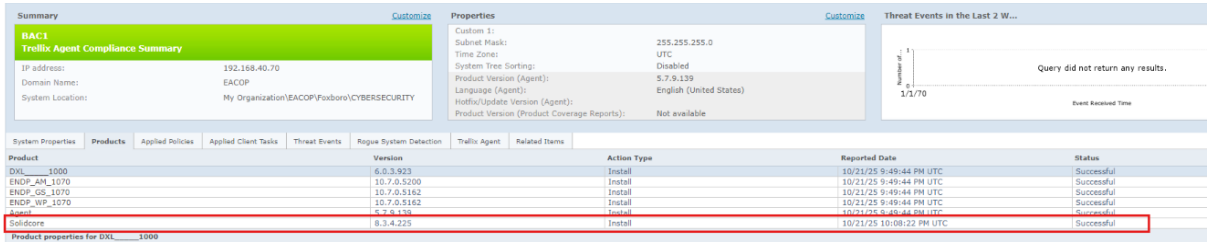
**Note:** Application Control and Change Control features could not be simulated in this project due to license limitations. Only Integrity Control was available and implemented.

For demonstration we will create a Folder on a BAC1 Server which has important configuration files for Foxboro DCS which we will monitor.

The file path is located at *C:\ProgramData\TestLab*

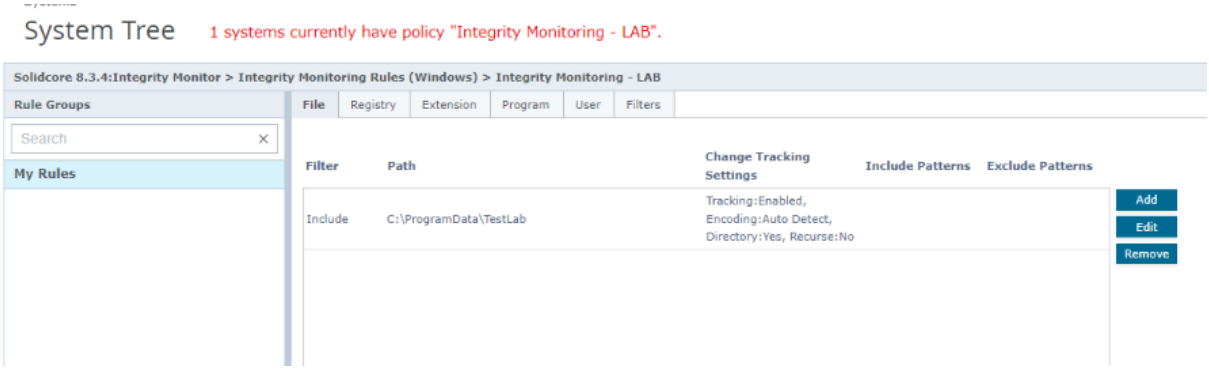
**Lab Steps**

1. Install Solidcore to a BAC1 Server

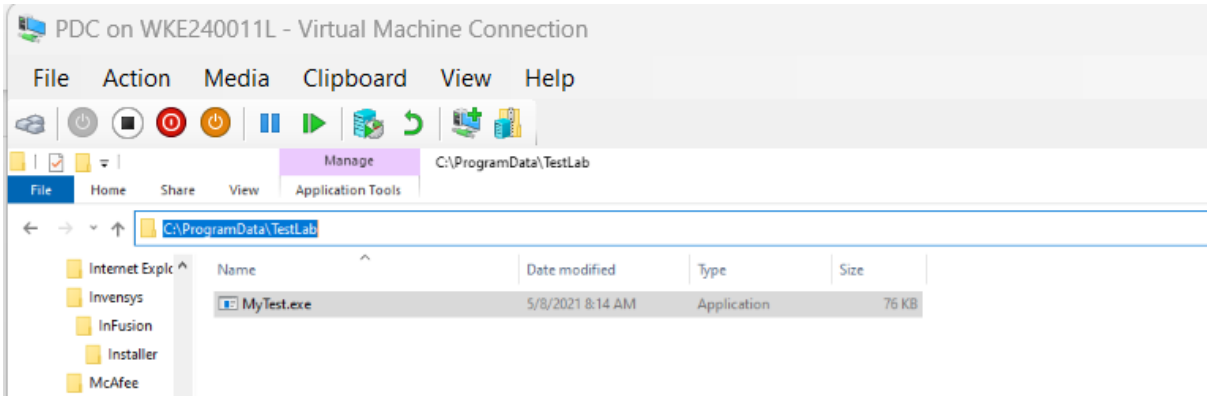



2. Enable Solicore on BAC1 Server

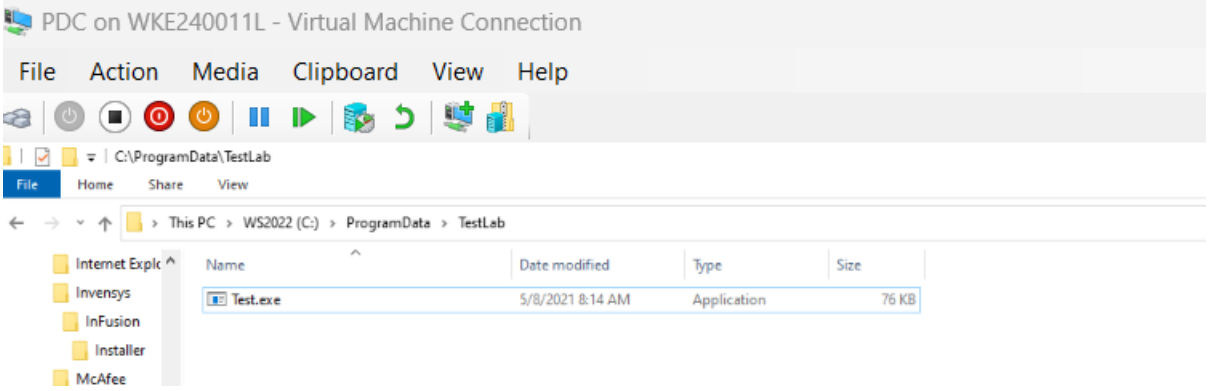
3. Create the Policy to monitor the folder located at *C:\ProgramData\TestLab*




4. Add the file in a monitored folder located at *C:\ProgramData\TestLab* on a PDC.



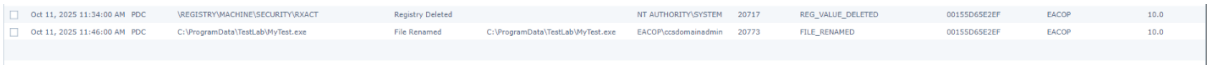

5. Generate a Custom Report.
6. Rename the file to **Test.exe**.




Observe the change reflected on a report.



7. Delete the file and observe the event is captured on a report.

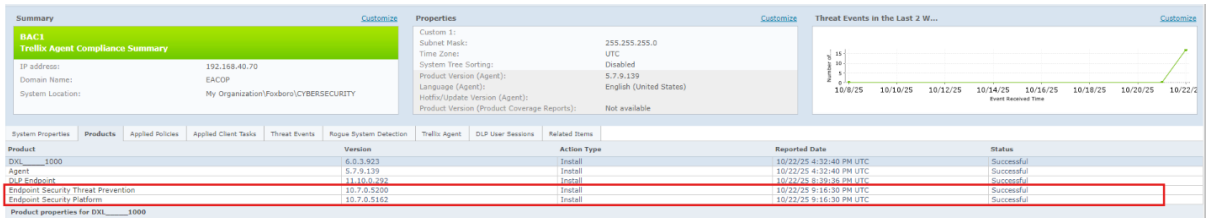


### *5.3.3 Threat Prevention*

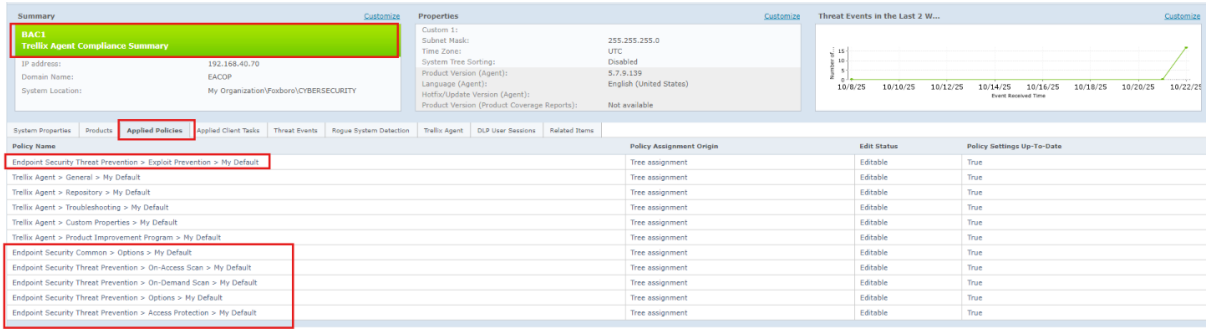
This prevents malware, ransomware, and malicious script from executing or scripting within OT System.

**Lab Steps**

3. From EPO Server install Threat Prevention to BAC1




4. Create a Policy for threat Prevention




5. Using Testing Malware/Ransomware Simulator to test the Threat Prevention.

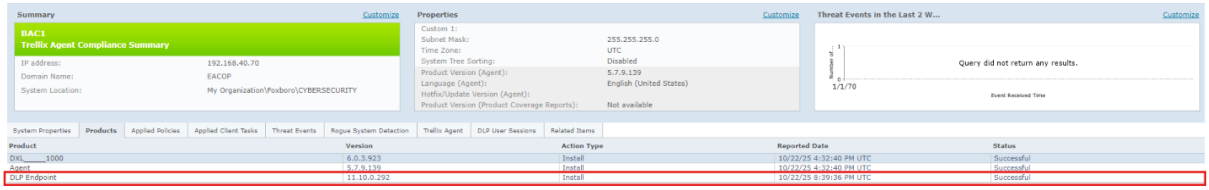
**Note**: Due to security restrictions on the host system running the virtual machines, this test could not be performed.

### *5.3.4 Device Control*

To Prevent unauthorized or malicious devices, such as USB drives, external disks, and removable media from being used to introduce malware or extract sensitive information within the OT cybersecurity lab.

**Lab Steps**

1. From EPO Server install Data Loss Prevention (DLP) Endpoint to BAC1




**Note:** Due to license limitations, the Device Control feature could not be implemented in this project.

## 5.4 R4 & R7 – Data Backup and Recovery

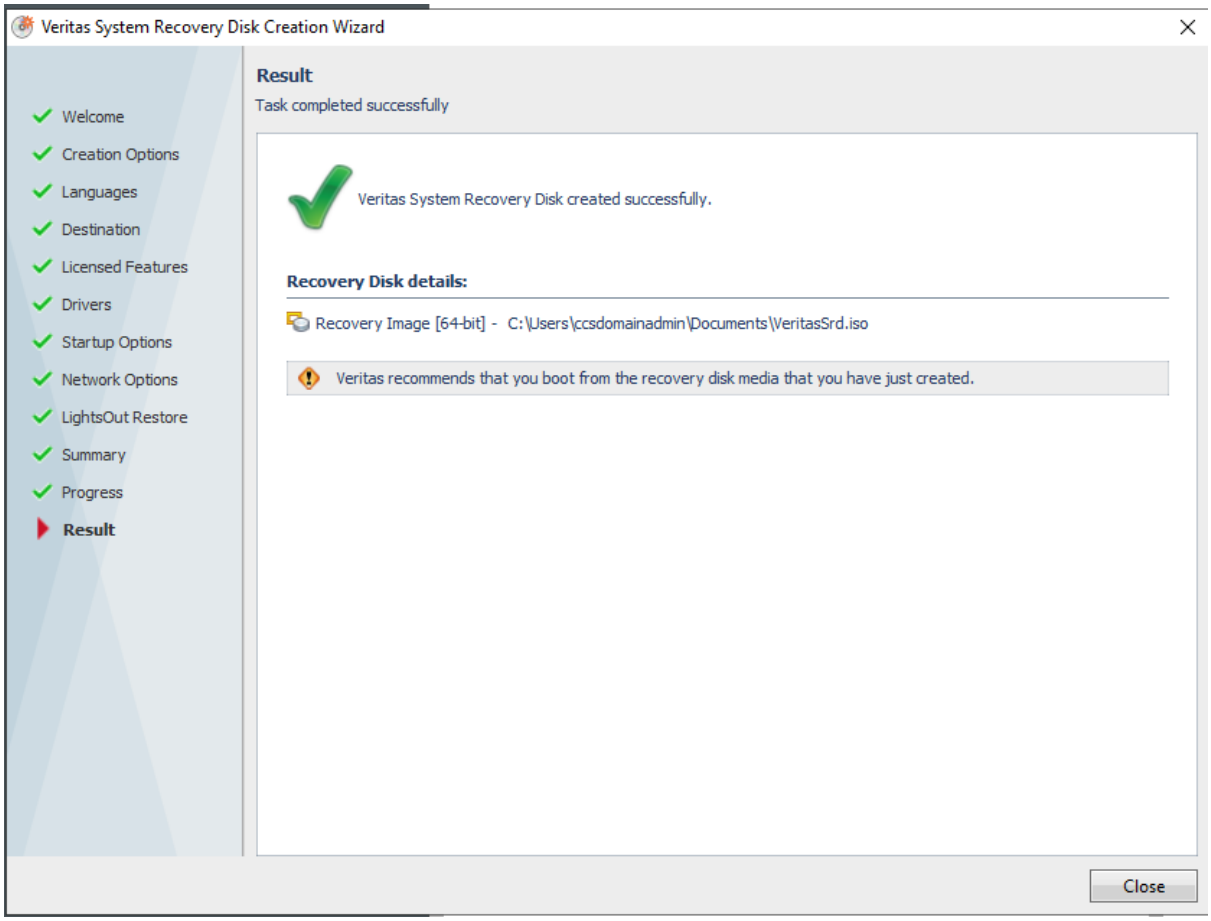
To ensure the availability and integrity of critical systems, configurations, and data within the OT environment, **Veritas System Recovery 23** is implemented as the primary backup and disaster recovery solution.

Key Actions:

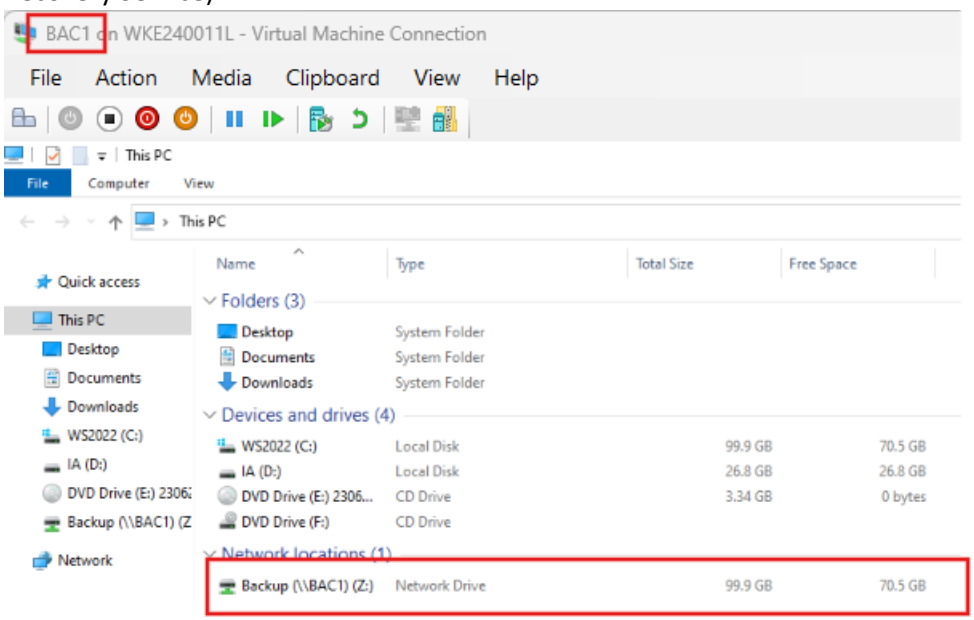
1. Create a Veritas System Recovery Disk for bootable recovery
2. Configure a shared network drive on the PDC server to serve as the backup repository
3. Schedule regular backups for critical drives and system configurations
4. Perform test restorations to validate backup integrity and recovery capability.

**Lab Steps**

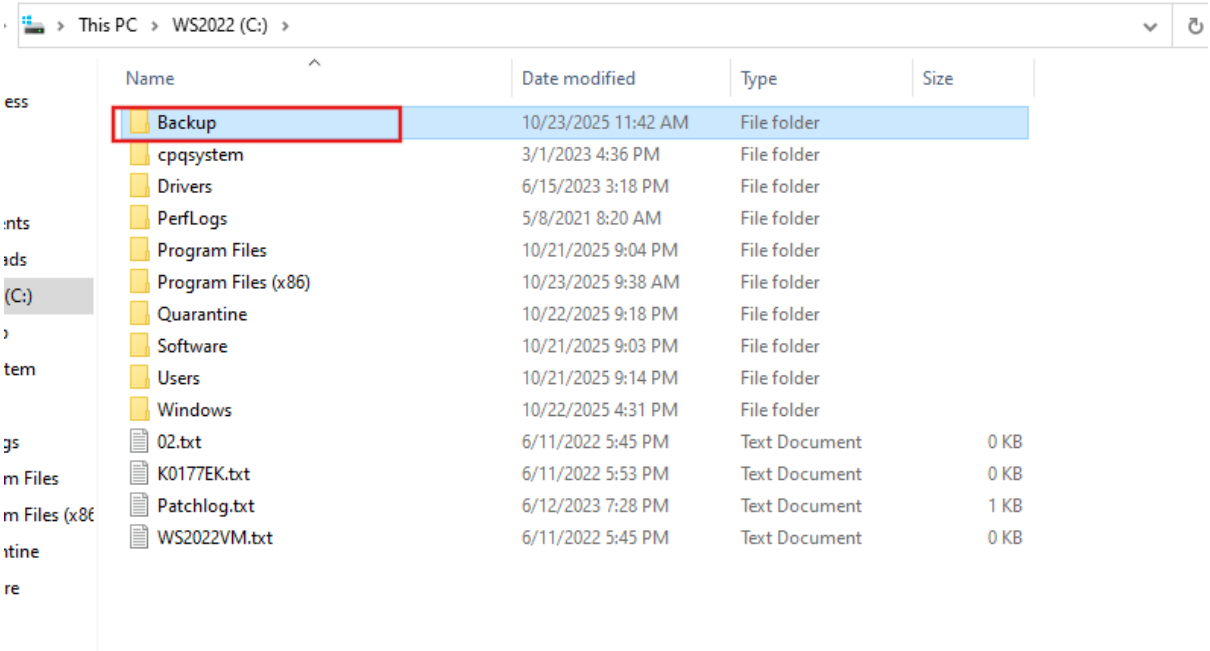
1. Create a Veritas System Recovery Disk



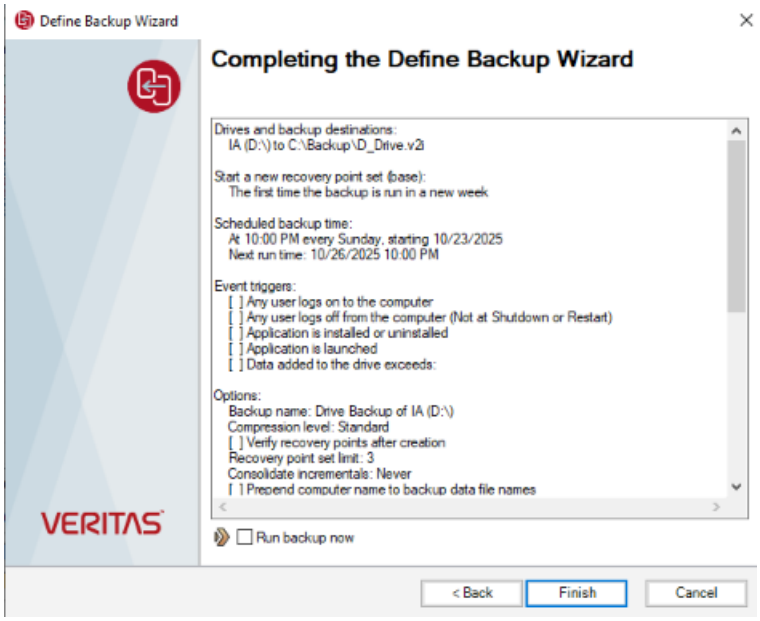

2. Setting up a shared Network Drive on PDC Server (This will be used for Managed Backup and Recovery Service)



For this lab, we will use a folder named **Backup** as the designated storage location for all backup files.

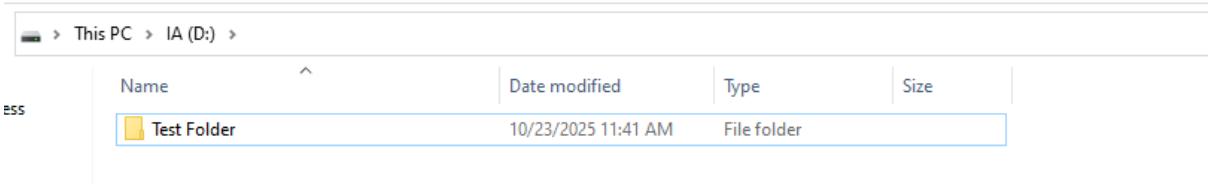



3. **Schedule a backup for Drive D** using Veritas System Recovery to ensure data protection and enable restoration in case of failure

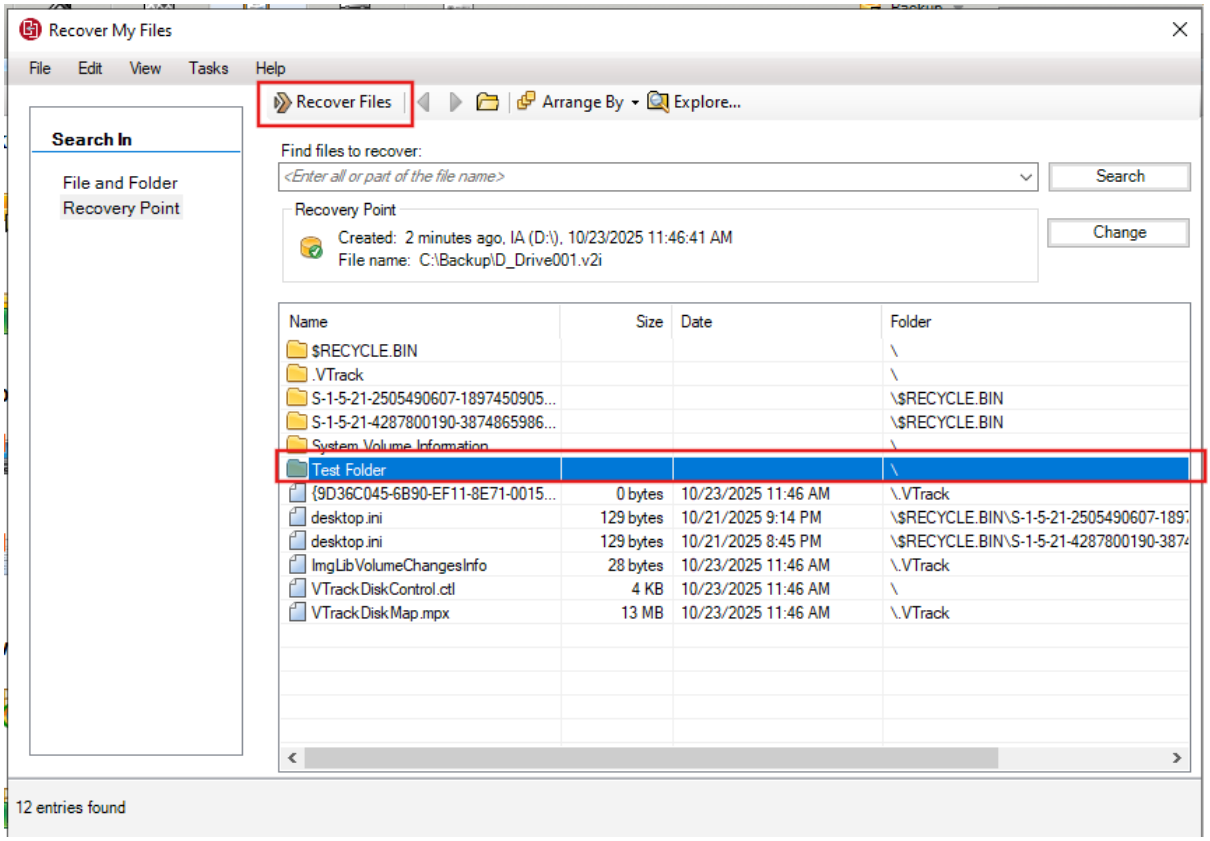



**Validation:**

1. Delete the **Test Folder** located on Drive D to simulate data loss and validate the backup restoration process.



2. Recover the deleted data using Veritas System Recovery to validate the backup and restoration process.



# Chapter 6: Conclusion

This project demonstrates key techniques for securing endpoints within an Operational Technology (OT) environment using Active Directory Domain Services, Trellix Endpoint Security solutions, and firewall-based network segmentation. While the scope was limited due to resource and licensing constraints, the implementation reflects a practical, hands-on understanding of cybersecurity principles aligned with industry standards such as ISA/IEC 62443 and NIST SP 800-30.

The lab successfully showcases how layered security controls can be applied to protect critical OT systems, validate risk mitigation strategies, and ensure resilience against cyber threats in industrial environments.